

Telemetry Anomaly Auditor

Submission Documentation

The `shutdown_triggered` flag is used to track whether a "STOP" termination signal has been detected during the audit. Initially, the flag is set to `False`, meaning that no shutdown signal has been found. While processing each telemetry reading, the program checks whether the reading is "STOP". If it is, the flag is changed to `True`, an emergency shutdown message is displayed, and the inner loop is terminated using `break`. After each batch, the outer loop checks the value of `shutdown_triggered`. If it is `True`, the outer loop also executes `break`, preventing its `else` clause from running. Python executes a `for` loop's `else` clause only when the loop completes normally without encountering `break`. Therefore, the `else` clause runs only when no "STOP" signal was detected anywhere in the telemetry stream.

Python Code

```
telemetry_stream = [
    [22.5, 23.0, 22.8],
    [25.1, "ERR", 24.9],
    [30.2, 35.5, 40.1], # Threshold breach
    [22.0, 22.1, "STOP"], # Termination signal
]

shutdown_triggered = False

for batch_id in range(len(telemetry_stream)):
    previous_value = None
    print(f"--- Auditing Batch {batch_id}: {telemetry_stream[batch_id]} ---")
    batches = telemetry_stream[batch_id]

    for reading in batches:
        if reading == "STOP":
            print(f"Emergency Shutdown at Batch {batch_id}.")
            shutdown_triggered = True
            break

        if reading == "ERR":
            print(f"Noise ignored at Batch {batch_id} (ERR).")
            continue
```

```

    if isinstance(reading, (int, float)):
        if reading > 35.0:
            print("Anomaly Detected at Batch", batch_id, ":", reading)

        if previous_value is not None:
            delta = abs(reading - previous_value)

            if delta > 5.0:
                print(
                    f"Spike detected at {batch_id}: "
                    f"{previous_value} -> {reading} "
                    f"Delta {delta:.1f}"
                )

            previous_value = reading

    if shutdown_triggered:
        break

else:
    print("Audit Complete : No system-wide failures")

```

Program Output

```

--- Auditing Batch 0: [22.5, 23.0, 22.8] ---
--- Auditing Batch 1: [25.1, 'ERR', 24.9] ---
Noise ignored at Batch 1 (ERR).
--- Auditing Batch 2: [30.2, 35.5, 40.1] ---
Anomaly Detected at Batch 2 : 35.5
Spike detected at 2: 30.2 -> 35.5 Delta 5.3
Anomaly Detected at Batch 2 : 40.1
--- Auditing Batch 3: [22.0, 22.1, 'STOP'] ---
Emergency Shutdown at Batch 3.

```

Conclusion

Since the telemetry stream contains a "STOP" signal in Batch 3, the `shutdown_triggered` flag becomes `True`. The outer loop then executes `break`, so the outer `else` clause is not executed.